AI Application R&D Engineer · BenQ Materials

# Shan-Cheng Chang 張善程

LLM Agent Engineering · Knowledge Graph · Document AI


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## — CORE SKILLS

**Agent Engineering**

Agent Harness · Function Calling · MCP · Skills / Tools · Context Management · Sandboxing · RBAC / Audit Log

**Evaluation & Validation**

Golden Set / Held-out · Code Validators · Regression Baseline · Failure Analysis

**LLM & Retrieval**

Gemini API · Azure OpenAI · Prompt Engineering · RAG · Knowledge Graph (Neo4j) · pgvector

**Document AI / ML / CV**

Azure Document Intelligence · OCR + LLM Extraction · YOLOv8 · PyTorch

**Backend & Web**

Python · FastAPI · Node.js · React · TypeScript · PostgreSQL · SQL Server

**DevOps & Tools**

Docker · GCP · Azure · Git · Claude Code

**Security**

AES-256-GCM · JWT · AD / Entra ID · ISO 27001

## — EDUCATION

**National Taiwan University**
M.S. Biomechatronics Eng. (AIoT / CV) · 2021 – 2023

**National Taipei University of Technology**
B.S. Mechanical Engineering · 2017 – 2021

AI application engineer who designs, validates and maintains enterprise AI systems: architect of BenQ Materials' AI data-analysis platform (in daily production use) and of its document-AI stack, built from zero. Every system is validated against a test set or the system it replaces before go-live.

| 4.15M NTD | 6 | ~1M NTD | 1,500+ h |
|---|---|---|---|
| AUTOML AVOIDED / 2 YRS | LINE FLOWS ON PLATFORM | BENEFIT / YR (3 OCR) | SAVED / YR (3 OCR) |

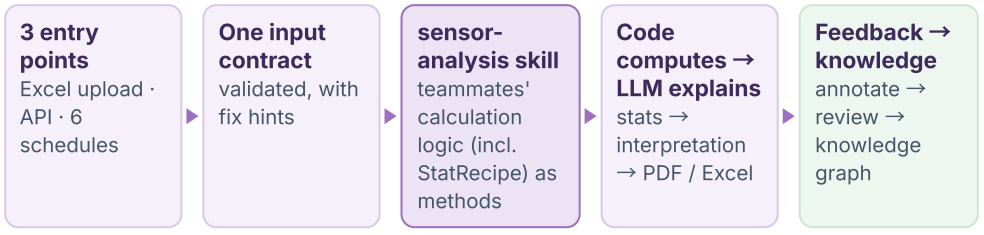
## — EXPERIENCE

### AI Application R&D Engineer

Dec 2024 – Present

BenQ Materials Corp. (明基材料) · Digital Technology Center, Intelligent Application Dept., Intelligent Application Development Sec. · Taoyuan, Taiwan

- **Enterprise Brain → AI data-analysis platform** (system architect & lead developer) — used daily by the BML line, replacing a purchased AutoML package (**4.15M NTD / 2 yrs**); 6 flows cut over with zero downtime. FastAPI + Gemini + MCP, on-prem.




Data → analysis → knowledge. Purple box: the shared skill I designed.

| | Before (n8n) | After (one skill) |
|---|---|---|
| Analysis logic | 6 workflows, ~550 nodes, each a copy | **1 skill shared by all 3 entry points** |
| New analysis method | edit every workflow | **teammate gives the logic; wired in once** |
| New data source | clone a whole workflow | **map data to the contract (7 sources)** |
| StatRecipe (SR) | a separate node chain | **ported as one skill method** |
| Cutover check | — | **Excel 9/9 cell-identical to old emails; identical PDFs** |

- **Design** — teammates supply only the calculation; the platform handles validation, interpretation and reports; an LLM failure never blocks the numbers.
- **Reliability & safety** — long chats auto-summarized, failed tools retried with the error fed back, max 10 tool calls per question; role-based access, rules enforced in code, tamper-proof audit log.
- **Knowledge loop** — the graph holds the 6 OCAP defect-handling flows I built; chat Q&A opens once it matures.

## — LANGUAGES

Mandarin Chinese — Native
English — Professional (TOEIC 790)
German — A2

## — CERTIFICATIONS

Company Digital Talent L3 (Expert)
Military service completed

## — EARLIER PROJECT

**Autonomous navigation with Autoware / ROS**
B.S. capstone, NTUT · 2020 – 2021
Indoor/outdoor self-driving platform (mapping, localization, path planning) with an ML mask-detection module.

## — PUBLICATIONS & PATENT

Chou, C.-Y., **Chang, S.-C.**, et al. (2023). Development of AIoT system for facility asparagus cultivation. *Computers and Electronics in Agriculture*, 206:107665. *(co-first author with advisor)*

**Chang, S.-C.**, et al. (2022). AIoT based pest counting system for asparagus cultivation. *Proc. 10th ISMAB*.

Taiwan Invention Patent TW I842416 B — Crop Harvester, 2024.

## — EXPERIENCE (CONT.)

### BenQ Materials — Document AI & other systems

Dec 2024 – Present

- **Document-AI applications (OCR + LLM)** — built the company's platform from zero (Azure Document Intelligence + Azure OpenAI); every system scored on a hand-labelled test set before go-live.

| System | Accuracy (test set) | Impact |
|---|---|---|
| **Quotation OCR** | validity date **51% → 100%** (29 docs) | 30 s → 10 s per doc; 758+ processed |
| **TW / CN invoice OCR** | **31/32**, **31/31** docs fully correct (incl. handwritten) | 1,447 h/yr saved; 3 sites |
| **Health-check OCR** | **97.4%** field-level (42 × 46 fields) | AES-256-GCM, AD login, ISO 27001 |
| **APO-OCR prompt tuning** | **8/8** held-out (24 train / 8 held-out) | $1.44; mentored an intern to ship it |

Scoring is exact-match code, the LLM only proposes; a 7/8 APO-OCR run traced to a date format missing from training.

- **Google Cloud Platform — supplier questionnaire management system** (live) — 52 APIs, built solo in 4.5 months, handed over; React 19 + Cloud Functions + Entra ID.
- **ESG carbon platform refactor & handover** — import moved to a web upload run by a colleague; backend split into one module per page (APIs 104 → 42), 55 copy-pasted chart files merged into 5 shared components; every step verified output-identical.

### Shift Supervisor (值班股長)

Jan 2024 – Nov 2024

China Steel Corporation (中國鋼鐵) · Cold-Rolling Mill · Kaohsiung, Taiwan

- **Ran shift operations** on the cold-rolled coil line at **95% yield**.
- **Anomaly analysis** — SPC, trend and correlation analysis for root causes of defects and downtime; daily anomaly reports (Python / Excel).

## — RESEARCH & SELECTED PROJECTS

### AIoT Pest Monitoring & Bee Pollination (M.S. thesis, NTU)

2021 – 2023

Lead developer of greenhouse AIoT systems: bee tracking *99.5% mAP*, pest counting *95.8%* (YOLOv8, StrongSORT, Mask R-CNN).

### Universal OCR Platform — no-code OCR builder (side project)

2025 – 2026

Define fields and prompts, deploy as an OCR REST API in one click (FastAPI · React · Cloud Run).